Modélisation, estimation, simulation des risques climatiques
Session 6

IAM Controversies

Gauthier Vermandel
Ecole polytechnique

IAMs in practice: evaluating the social cost of carbon

- IAMs are not only academic tools: since 2010, the US federal government uses them to compute the **social cost of carbon** (SCC), the damage caused by one extra ton of CO_2.
- An interagency working group averages three IAMs (DICE, PAGE, FUND); revisions across administrations show how normative choices move the number.

Administration	SCC ($/tCO$_2$)	Main change
Obama (2010)	21	IWG, 3% discount rate
Obama (2016)	42	updated DICE/PAGE/FUND runs
Trump (2017–2021)	1–7	domestic damages, 7% rate
Biden (2021)	51	interim, global damages
EPA (2023)	190	updated science, Ramsey 2%

How relevant are IAM models?

- **IAM-believiers:** they are relevant at least on normative grounds to provide discussion basis on mitigation scenarios.

- **IAM-deniers:** they are useless to guide policy scenarios → the Pindyck (2013, 2017) critique enumerates a list of flaws:

  1. "*certain inputs [parameters] are arbitrary, but have huge effects on the SCC estimates the models produce*";
  2. "*the models' descriptions of the impact of climate change are completely ad hoc, with no theoretical or empirical foundation*";
  3. "*the models can tell us nothing about the most important driver of the SCC, the possibility of a catastrophic climate outcome*"

Two debates behind the critiques of IAMs

- IAMs leave many degrees of freedom: the same framework can support very different climate policies → an impression of lack of discipline.

- Two parameters illustrate this paradox:

  1. **Discounting:** the Stern–Nordhaus debate. How much weight should future damages receive relative to mitigation costs today?

  2. **Damage functions:** still an ongoing debate, on two fronts (treated in the sections below):

     2.1 **functional form:** how large are climate damages, and how should abrupt, irreversible losses be represented?

     2.2 **temperature elasticity:** how fast do damages rise with warming — quadratic in T, or higher-order terms?

The Discount Factor Controversy

The Discount Factor Controversy

- In 2006, the main view on climate change among economists was the Nordhaus (1992)'s view: optimal to warm up to $3.5°C$ the planet.
- The Stern (2006)'s view shook this general agreement on many aspects:
  - the core argument: the price of inaction would be extraordinary and the cost of action modest ;
  - irreversible impacts from climate change threatening access to water, food production, health, and use of land and the environment;
  - IAM supports such actions in particular by changing discount factor → debate.



Nordhaus Stern

Photos: B. Nyman, CC BY 2.0; IMF

The Discount Factor Controversy

- What's discount factor? Welfare function as a discounted sum of future gains:

$$\mathcal{W}_t = L_t U\left(c_t\right) + \beta_\Delta \mathcal{W}_{t+\Delta}$$

- β_Δ is a reduced-form parameter: a single number summarizing several distinct motives for weighting the future:

  1. psychological aspects (time-preference $\rightarrow \beta_\Delta$ low),
  2. political aspects (how policymakers weight the future $\rightarrow \beta_\Delta$ low/high),
  3. intergenerational aspects (how we weight future generation welfare $\rightarrow \beta_\Delta$ high).

The Discount Factor Controversy

- In practice, β_Δ not observable, but can be inferred from financial markets.
- Introducing a one-period financial claim:

$$\max_{\{c_t, b_t\}} \sum_{i=0}^{\infty} \beta_\Delta^i \frac{c_{t+\Delta i}^{1-\gamma}}{1-\gamma} + \lambda_t \ \left[b_t \left(1 + r_{t-\Delta}\right) - c_t - b_t\right]$$

$$\beta \frac{c_{t+\Delta}^{-\gamma}}{c_t^{-\gamma}} \left(1 + r_t\right) = 1 \rightarrow \ln \beta_\Delta = \gamma \ln g + \ln r$$

- By observing consumption growth and real interest rate, we can pin down β_Δ;
- But definition of relevant interest rate not clear: nominal bonds? corporate bonds?

The Discount Factor Controversy (cont.)

- Nordhaus picks the market return on capital (corporate rates, about 3% real) and calibrates $\rho = 1.5\%$, i.e. $\beta_{\Delta=1} = 1/1.015 \simeq 0.985$;
- Stern (2006) argues that discount factor should be $\boxed{\beta_{\Delta=1} = 0.999}$ (versus 0.9852 in Nordhaus).

Nordhausś calibration of ρ across DICE vintages:

	ρ	$\beta_{\Delta=1} = 1/(1+\rho)$
DICE-1994	3.0%	0.9709
DICE-2007–2016	1.5%	0.9852
DICE-2023	1.0%	0.9901
Stern (2006), for contrast	0.1%	0.9990

The discount factor

- The Stern (2006)'s report provides a scientific motivation for the Paris-Agreement.
- How? Consider FOC on T:

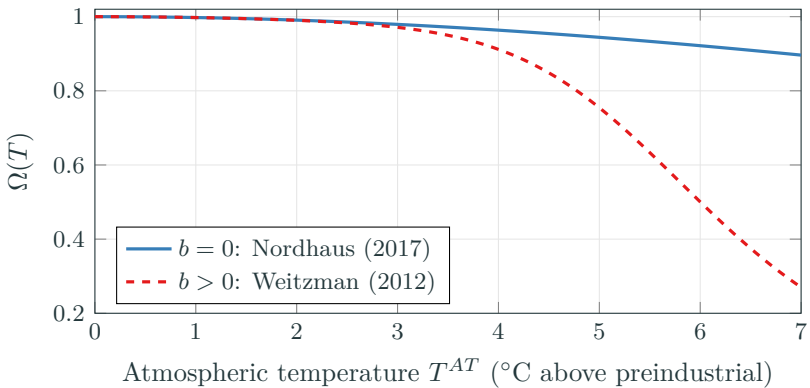
$$\begin{aligned}\lambda_{4,t} &= \beta_\Delta \left[\Phi_T^\top \lambda_{4,t+\Delta} + \lambda_{2,t+\Delta} s_{t+\Delta} Y_{t+\Delta} \left(1 - \theta_{1,t+\Delta} \mu_{t+\Delta}^{\theta_2}\right) \nabla_T \Omega\left(T_t\right)\right] \\ &= \sum_{i=1}^{\infty} \left(\beta_\Delta \Phi_T^\top\right)^i s_{t+\Delta i} Y_{t+\Delta i} \left(1 - \theta_{1,t+\Delta i} \mu_{t+\Delta i}^{\theta_2}\right) \nabla_T \Omega\left(T_{t+\Delta(i-1)}\right)\end{aligned}$$

- Bigger β raises the weight β_Δ^i on future marginal damages $\nabla_T \Omega$ in the shadow value of temperature $\lambda_{4,t} \rightarrow$ larger weight for future generations, SCC $\nearrow$

The Damage Function and Tipping Points

Damage functions: Nordhaus vs. Weitzman

Share of output left after climate damages, $\Omega(T)$, up to 7°C:



Tipping points

- Tipping point are critical threshold beyond which a system reorganizes abruptly & irreversibly.
- Natural system tipping points well identified:
  - Antarctic & Artic ice sheet disintegration: complete disintegration (at +10°C) would raise the global sea levels by 53.3 metres;
  - Amazon Rainforest destruction: a warming planet →rainforest may transform into a dry savanna landscape;
  - Sibera's Permafrost thaw: unfreezing permafrost would release methane ↗ T°C.
  - etc.
- Tipping points are possible at today's global warming of just over 1 °C (1.8 °F) above preindustrial times, and highly probable above 2 °C (3.6 °F) of global warming.

Tipping points

- Tipping point are critical threshold beyond which a system reorganizes abruptly & irreversibly.
- The damage function in Nordhaus (2017) augmented by Weitzman (2012) with higher order polynomial:

$$\Omega\left(T_t\right) = 1/(1 + a_2 \left(T_t^{AT}\right)^2 + bT_t^c)$$

where b and c are set to match potential losses as T grows: $b = 5.0703e - 06$, $c = 6.754$.

	$\Omega\left(1\right)$	$\Omega\left(3\right)$	$\Omega\left(5\right)$
$b = 0$ - Nordhaus (2017)	0.9976	0.9792	0.9443
$b > 0$ - Weitzman (2012)	0.9976	0.9712	0.7544

Tipping points

- Why tipping points call for stronger actions?
- Get back to FOC:

$$\lambda_{4,t}^{AT} = \sum_{i=1}^{\infty} \left(\beta_{\Delta} \Phi_T^{\top}\right)^i s_{t+\Delta i} Y_{t+\Delta i} \left(1 - \theta_{1,t+\Delta i} \mu_{t+\Delta i}^{\theta_2}\right) \nabla_T \Omega \left(T_{t+\Delta(i-1)}\right)$$

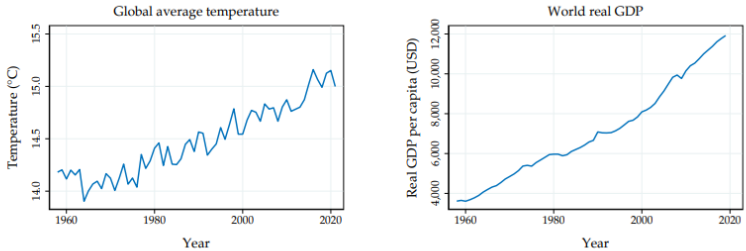
- $\nabla_T \Omega \left(T_{t+\Delta(i-1)}\right)$ much larger under tipping climate, thus rising the Social Cost of Carbon and in turn the carbon tax.

Estimating damages: global vs. local temperature

Empirical damages: global vs. local temperature

- The damage function is the weakest link of IAMs: a_2 is calibrated, not estimated $\rightarrow$ can we **estimate** it from macro data?
- Bilal and Känzig (2026): shift the focus from **local** to **global** temperature:
  1. a 1°C global warming lowers world GDP by **over 20%** in the long run, an order of magnitude more than with country-level temperature;
  2. global temperature correlates strongly with **extreme events**, country-level temperature does not.
- Feeding the estimated damage function into a neoclassical growth model: business-as-usual welfare loss **above 30%**, SCC **above \$1 200/tCO$_2$**.

Global temperature and world GDP: two trending series



Notes: Global average temperature (including sea surface) from NOAA, world real GDP from PWT

- Both global temperature and world GDP **trend upward**.
- Regressing one on the other → risk of **spurious correlation**.
- Solution: focus on **temperature shocks**, i.e. transitory deviations around the trend.

Measuring temperature shocks

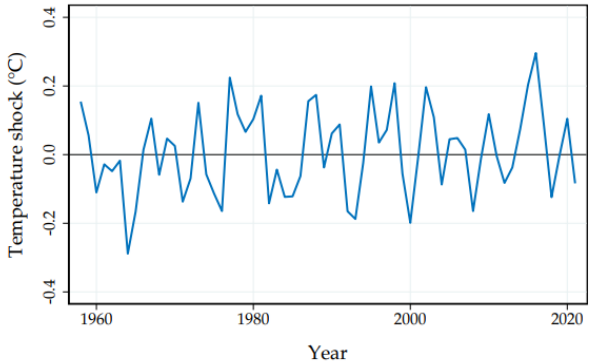
- Following **Hamilton (2018)**: the shock is the **forecast error** of temperature:

$$\widehat{T}^{\text{shock}}_{t+h} = T_{t+h} - \left(\hat{\beta}_0 + \hat{\beta}_1 T_t + \cdots + \hat{\beta}_p T_{t-p+1}\right)$$

- Sources of variation around the trend:
  1. exogenous drivers: solar cycles, volcanic eruptions;
  2. internal climate variability: ocean circulation, El Niño.

- $h = 2$, $p = 2$ to capture **persistent** climate phenomena (e.g. El Niño); results robust to other specifications.

Exogenous temperature shocks



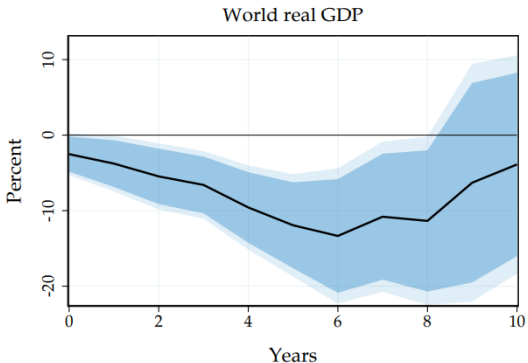
Estimating the effects of global temperature shocks

- Dynamic causal effects estimated by **local projections** (Jordà, 2005):

$$y_{t+h} - y_{t-1} = \alpha_h + \theta_h T_t^{\text{shock}} + \mathbf{x}_t' \boldsymbol{\beta}_h + \varepsilon_{t+h}$$

- y_t: log real world GDP per capita; T_t^{shock}: global temperature shock; $\mathbf{x}_t$: controls.
- θ_h: **causal effect at horizon** $h \rightarrow$ the sequence $\{\theta_h\}$ traces the impulse response of GDP.

Effects of global temperature shocks



Notes: 90% and 95% confidence bands, robust standard errors.

- Significant and **persistent** effect.
- After a 1°C shock:
  1. GDP per capita falls by **2%** on impact;
  2. the effect exceeds **10%** after 6 years;
  3. still there 10 years later, driven by the **persistence of temperature** itself.

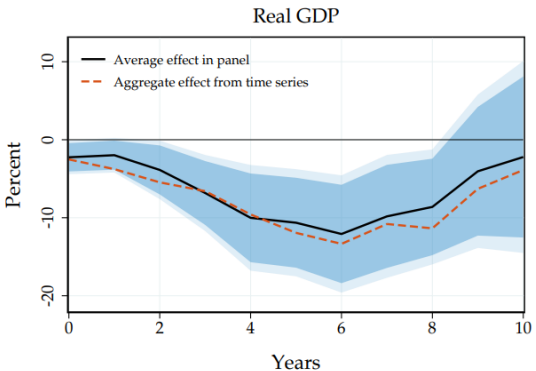
A climate–economy panel

- **173 countries**, from 1960 (back to 1900 for some).
- Economic data: PWT and JST Macrohistory (GDP, population, capital, investment, productivity); temperature: Berkeley Earth; extreme events: ISIMIP gridded data.
- **Panel local projections** (Jordà et al., 2020):

$$y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \theta_h T_t^{\text{shock}} + \mathbf{x}_t' \boldsymbol{\beta}_h + \mathbf{x}_{i,t}' \boldsymbol{\gamma}_h + \varepsilon_{i,t+h}$$

- Allows comparing responses to **global** and **local** temperature shocks.

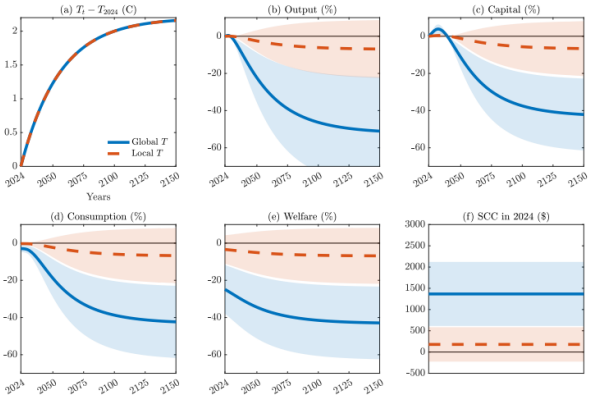
Global temperature shocks in the panel



Notes: 90% and 95% bands, Driscoll–Kraay standard errors.

- **Substantial** impact: GDP per capita falls by more than **10%**.
- Panel effect $\approx$ time-series effect.
- Local temperature shocks: much smaller effect, in line with the earlier literature.

Implications for IAMs



Notes: simulated impact of global vs. local temperature damages, model calibrated in 2024.

- **Global** damages: in 2100, C and Y -40%; welfare loss 25%; SCC $= \$1\,367/\text{tCO}_2$.
- **Local** damages: C and Y -6%; welfare loss 3%; SCC $= \$178/\text{tCO}_2$, close to standard IAMs.
- Same model, different **damage evidence** $\rightarrow$ SCC $\times 7$: aggressive decarbonization becomes optimal.

Recap

Exam Guidelines

- **Equations:** You don't need to memorize every equation by heart. But you should be able to recognize them if you see them, and explain their meaning.
- **Simulations:** You don't need to know the code. What matters is that you understand the outputs (graphs, tables, numbers) and can interpret them correctly.
- **MCQs:** Expect multiple-choice questions with text, equations, and figures. The goal is to test whether you can read, interpret, and choose the correct answer.

In short: focus on understanding, not rote learning.

Thank you!

gauthier@vermandel.fr

References

Bilal, A. and Känzig, D. R. (2026). The macroeconomic impact of climate change: Global versus local temperature. *The Quarterly Journal of Economics*, 141(2):889–944.

Nordhaus, W. (1992). The 'DICE' model: Background and structure of a dynamic integrated climate-economy model of the economics of global warming. Technical report, Cowles Foundation for Research in Economics, Yale University.

Nordhaus, W. (2017). Revisiting the social cost of carbon.

Pindyck, R. S. (2013). Climate change policy: what do the models tell us? *Journal of Economic Literature*, 51(3):860–872.

Pindyck, R. S. (2017). The use and misuse of models for climate policy. *Review of Environmental Economics and Policy*.

Stern, N. (2006). Stern review: The economics of climate change.

Weitzman, M. L. (2012). Ghg targets as insurance against catastrophic climate damages. *Journal of Public Economic Theory*, 14(2):221–244.